CSE 464: Software QA and Testing
Project Part 2 - README

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Github Repository: https://github.com/devjanyani/CSE-464-2026-djanyani

How to Build and Test

Clone the repository and run the following command from the **graph-parser** directory:

```
cd graph-parser
mvn package
```

This will compile the code, run all unit tests, and package the JAR file. Java JDK 25 and Maven are required.

Feature 1: Remove Node/Edge APIs

Three new APIs were added to the Graph class to support removing nodes and edges:

removeNode(String label) - Removes a node and all edges connected to it. Throws IllegalArgumentException if the node does not exist.

removeNodes(String[] labels) - Removes multiple nodes. Throws IllegalArgumentException if any node does not exist.

removeEdge(String srcLabel, String dstLabel) - Removes a specific edge. Throws IllegalArgumentException if the source node, destination node, or edge does not exist.

Example Usage:

```
Graph g = DotParser.parseGraph("input/sample.dot");
System.out.println(g.toString());

g.removeNode("a");
System.out.println(g.toString());

g.removeEdge("e", "f");
System.out.println(g.toString());
```

Screenshot - Remove API Output:

```
Run    Main
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\lib\idea_rt.jar=58455" -Dfile.encoding=
=== Parsed Graph ===
Number of nodes: 8
Nodes: [a, b, c, d, e, f, g, h]
Number of edges: 9
Edges:
a -> b
a -> e
b -> c
c -> d
d -> a
e -> f
e -> g
f -> h
g -> h

=== After removing node 'a' ===
Number of nodes: 7
Nodes: [b, c, d, e, f, g, h]
Number of edges: 6
Edges:
b -> c
c -> d
e -> f
e -> g
f -> h
g -> h

=== After removing edge 'e' -> 'f' ===
Number of nodes: 7
Nodes: [b, c, d, e, f, g, h]
Number of edges: 5
Edges:
b -> c
c -> d
e -> g
f -> h
g -> h
```

Commit:
https://github.com/devjanyani/CSE-464-2026-djanyani/commit/85523cc

Test Cases for Remove APIs

The following test scenarios are covered:

Scenario 1 - Correct Removal: Tests that nodes and edges are correctly removed, node/edge counts decrease, and edges involving removed nodes are cleaned up. Tests: testRemoveNode, testRemoveNodes, testRemoveEdge.

Scenario 2 - Removing Non-existent Nodes: Tests that removing a node that does not exist throws an IllegalArgumentException. Tests: testRemoveNodeNotExists, testRemoveNodesOneNotExists.

Scenario 3 - Removing Non-existent Edges: Tests that removing an edge that does not exist (or whose source/destination node does not exist) throws an IllegalArgumentException. Tests: testRemoveEdgeNotExists, testRemoveEdgeSrcNotExists.

Screenshot - All 17 Tests Passing (mvn package):

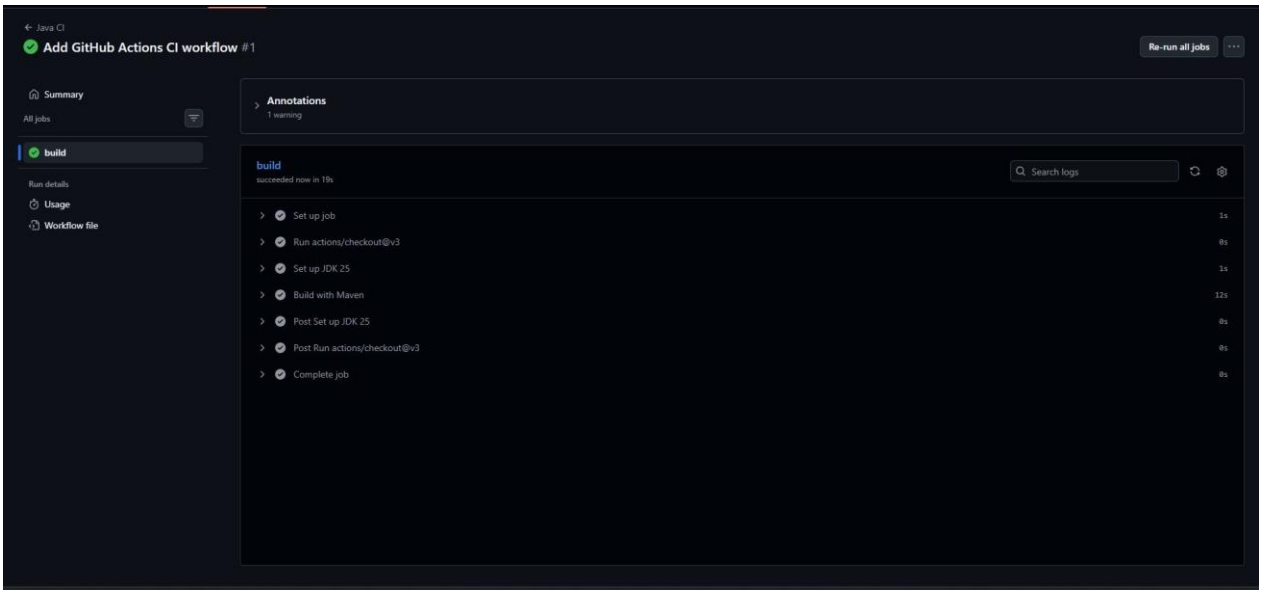
```
Terminal   Local

PS C:\Users\dejan\IdeaProjects\CSE-464-2026-djanyani\graph-parser> mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< edu.asu.cse464:graph-parser >---------------------
[INFO] Building graph-parser 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ graph-parser ---
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ graph-parser ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ graph-parser ---
[INFO] skip non existing resourceDirectory C:\Users\dejan\IdeaProjects\CSE-464-2026-djanyani\graph-parser\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ graph-parser ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ graph-parser ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running edu.asu.cse464.DotParserTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.085 s -- in edu.asu.cse464.DotParserTest
[INFO] Running edu.asu.cse464.GraphTest
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in edu.asu.cse464.GraphTest
[INFO] Running edu.asu.cse464.GraphvizExportTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.438 s -- in edu.asu.cse464.GraphvizExportTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 17, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ graph-parser ---
[INFO] Building jar: C:\Users\dejan\IdeaProjects\CSE-464-2026-djanyani\graph-parser\target\graph-parser-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.920 s
[INFO] Finished at: 2026-04-02T20:42:04-07:00
[INFO] ------------------------------------------------------------------------

PS C:\Users\dejan\IdeaProjects\CSE-464-2026-djanyani\graph-parser>
```

Feature 2: Continuous Integration (GitHub Actions):

Workflow Configuration:

```
name: Java CI
on: [push, pull_request]
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - name: Set up JDK 25
        uses: actions/setup-java@v3
        with:
          java-version: '25'
          distribution: 'temurin'
      - name: Build with Maven
        run: mvn -B package --file graph-parser/pom.xml
```

Screenshot - GitHub Actions CI Passing:



Commit:

https://github.com/devjanyani/CSE-464-2026-djanyani/commit/ea53fe9

CI Workflow Runs:

https://github.com/devjanyani/CSE-464-2026-djanyani/actions

Feature 3: BFS Graph Search (bfs branch):

API:

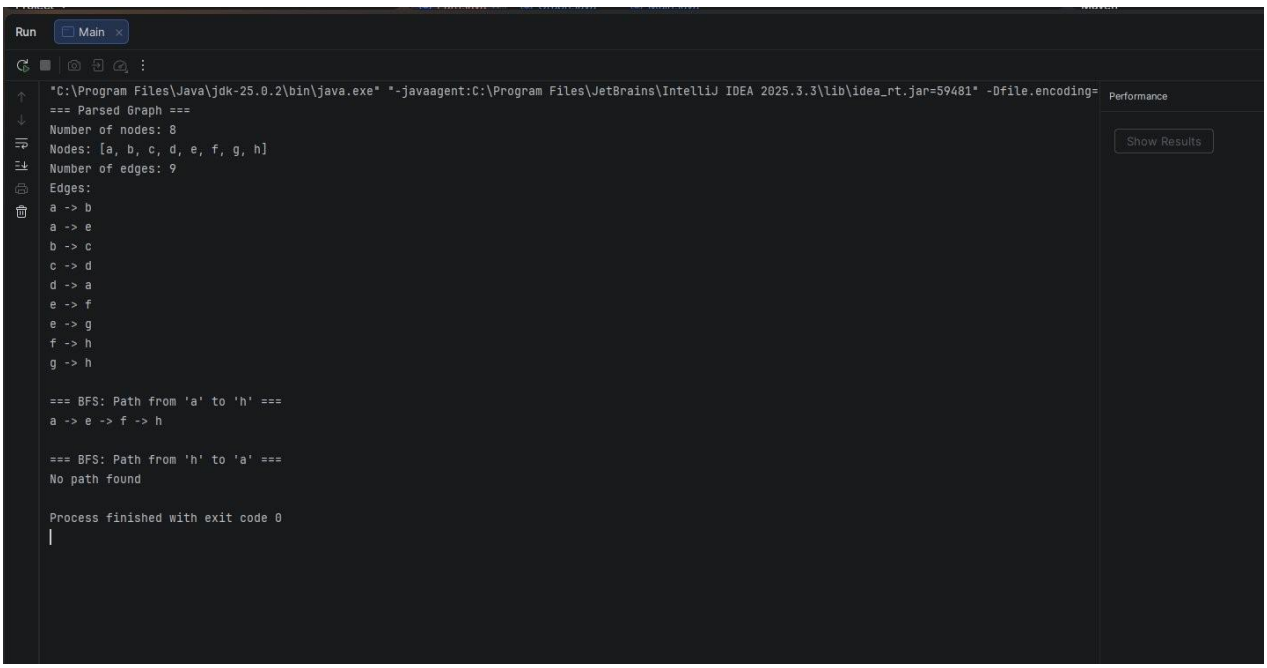
```
Path GraphSearch(String src, String dst)
```

This API finds a possible path from node src to node dst using BFS. Returns a Path object if a path exists, otherwise returns null.

Example Usage:

```
Graph g = DotParser.parseGraph("input/sample.dot");
Path path = g.GraphSearch("a", "h");
System.out.println(path); // a -> e -> f -> h
```

Screenshot - BFS Output:



Branch:

https://github.com/devjanyani/CSE-464-2026-djanyani/tree/bfs

Commit:

https://github.com/devjanyani/CSE-464-2026-djanyani/commit/12816c5

Feature 4: DFS Graph Search (dfs branch):

API:

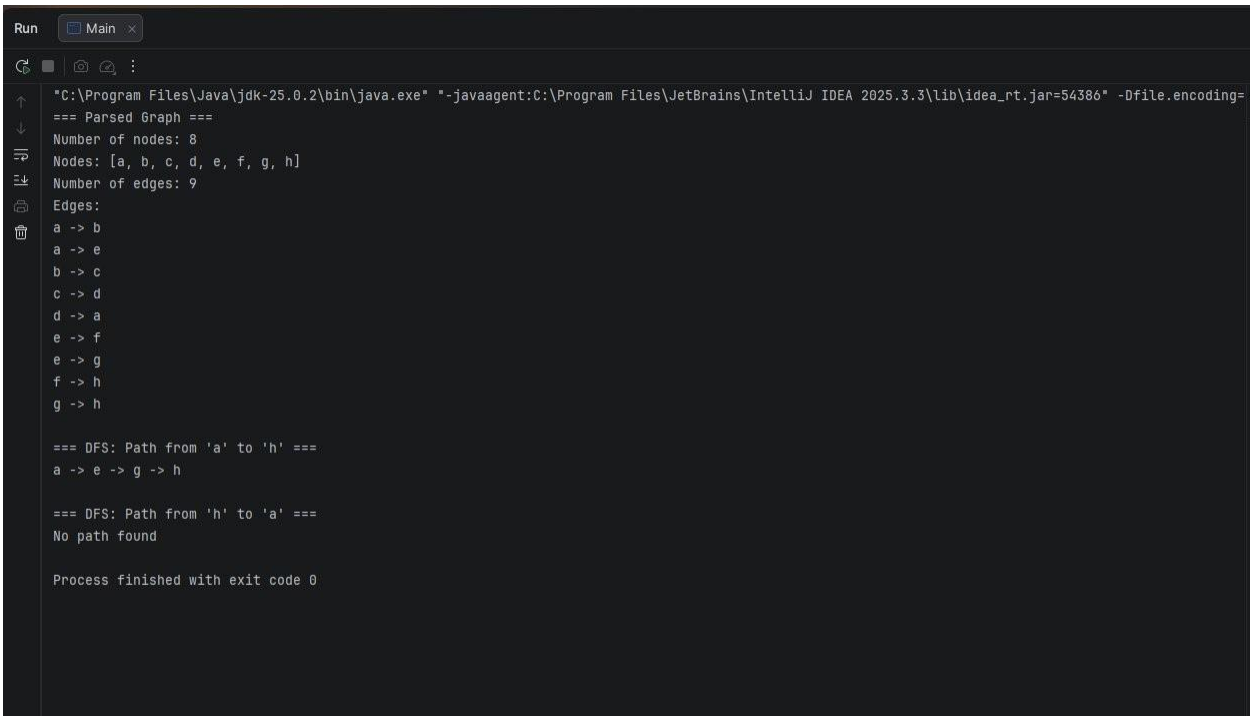
```
Path GraphSearch(String src, String dst)
```

This API finds a possible path from node src to node dst using DFS. Returns a Path object if a path exists, otherwise returns null. Note that DFS may find a different path than BFS.

Example Usage:

```
Graph g = DotParser.parseGraph("input/sample.dot");
Path path = g.GraphSearch("a", "h");
System.out.println(path); // a -> e -> g -> h
```

Screenshot - DFS Output:



Branch:

https://github.com/devjanyani/CSE-464-2026-djanyani/tree/dfs

Commit:

https://github.com/devjanyani/CSE-464-2026-djanyani/commit/63b3b1d

Feature 5: Merge and Algorithm Enum:

Algorithm Enum:

```
public enum Algorithm {
    BFS,
    DFS
}
```

Final Merged API:

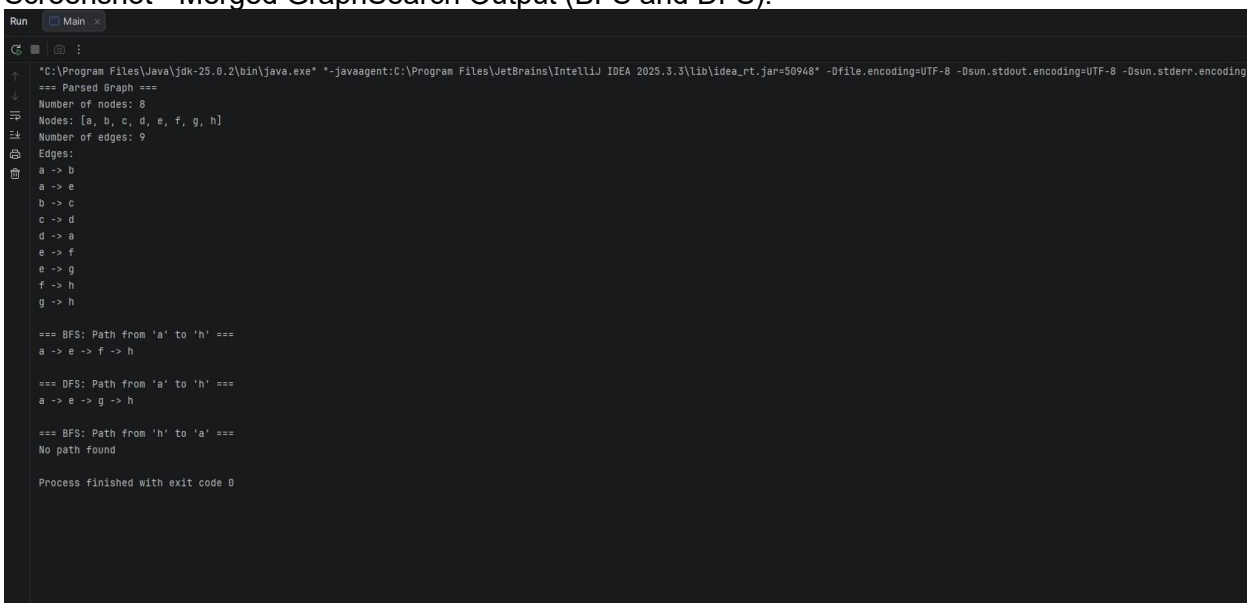
```
Path GraphSearch(String src, String dst, Algorithm algo)
```

This API finds a possible path from node src to node dst using the specified algorithm (either BFS or DFS). Returns a Path object if a path exists, otherwise returns null.

Example Usage:

```
Graph g = DotParser.parseGraph("input/sample.dot");

// BFS search
Path bfsPath = g.GraphSearch("a", "h", Algorithm.BFS);
System.out.println(bfsPath); // a -> e -> f -> h

// DFS search
Path dfsPath = g.GraphSearch("a", "h", Algorithm.DFS);
System.out.println(dfsPath); // a -> e -> g -> h
```

Screenshot - Merged GraphSearch Output (BFS and DFS):



Merge Commit:

https://github.com/devjanyani/CSE-464-2026-djanyani/commit/666e9d2

Summary of All Commits and Branches

f9cf846 (main): Fix duplicate edge check and add missing tests:

https://github.com/devjanyani/CSE-464-2026-djanyani/commit/f9cf846

85523cc (main): Add removeNode, removeNodes, removeEdge APIs with test cases

https://github.com/devjanyani/CSE-464-2026-djanyani/commit/85523cc

ea53fe9 (main): Add GitHub Actions CI workflow:

https://github.com/devjanyani/CSE-464-2026-djanyani/commit/ea53fe9

12816c5 (bfs): Add BFS GraphSearch with Path class:

https://github.com/devjanyani/CSE-464-2026-djanyani/commit/12816c5

63b3b1d (dfs): Add DFS GraphSearch with Path class:

https://github.com/devjanyani/CSE-464-2026-djanyani/commit/63b3b1d

666e9d2 (main): Merge bfs and dfs branches, resolve conflict with Algorithm enum

https://github.com/devjanyani/CSE-464-2026-djanyani/commit/666e9d2

Branch Links:

https://github.com/devjanyani/CSE-464-2026-djanyani/tree/bfs
https://github.com/devjanyani/CSE-464-2026-djanyani/tree/dfs
https://github.com/devjanyani/CSE-464-2026-djanyani/actions